

Facial Expression Recognition Using a Lightweight CNN on FER-2013 Dataset for Real-Time Emotion Detection

Abdullah Firoj*, Md. Rahamat Ullah[†], Apurba Kumar Paul[‡], Nurjahan Eliza[§], Sunjid Ferdous[¶]
 Dept. of Computer Science and Engineering, United International University, Dhaka, Bangladesh
 Emails: *afiroj221429@bscse.uiu.ac.bd, [†]mullah221325@bscse.uiu.ac.bd, [‡]apaul221320@bscse.uiu.ac.bd,
[§]neliza2310329@bscse.uiu.ac.bd, [¶]sferdous213091@bscse.uiu.ac.bd

Abstract—Facial expression recognition (FER) enables machines to interpret emotional states through visual cues. In this study, we present a compact Convolutional Neural Network (CNN) optimized for resource-limited environments such as embedded systems and mobile devices. Trained and validated on the FER-2013 dataset, our model achieves a test accuracy of 65%, balancing performance and efficiency with only 0.8 million parameters. We explore preprocessing techniques, extensive augmentation strategies, and deployment performance, achieving real-time inference at 12 FPS on mid-range smartphones using TensorFlow Lite. We compare with EfficientNet-B2 to elucidate trade-offs. Finally, we analyze misclassifications and propose future improvements.

Index Terms—Facial expression recognition, Lightweight CNN, FER-2013, Real-time inference, Emotion detection, Embedded vision, Deep learning deployment.



Fig. 1: Training/Validation Accuracy Loss Curves

I. INTRODUCTION

Facial expressions communicate rich emotional content non-verbally, making FER a cornerstone in affective computing and human-computer interaction (HCI). Applications range from mental health monitoring [?] and driver alertness systems [?] to intelligent tutoring and customer service analytics. Deep learning, particularly CNNs, has advanced FER accuracy significantly, yet model sizes and computational expenses often restrict deployment in low-power environments.

This research aims to design and analyze a lightweight CNN (0.8M parameters) that can operate in real time on constrained devices while delivering competitive accuracy. Our contributions include:

- A detailed, efficiency-driven CNN design.
- Systematic analysis of preprocessing, augmentation, and training strategies.
- On-device benchmarking on mobile hardware (12 FPS on Snapdragon 730).
- Comparison with EfficientNet-B2 and discussion on trade-offs.
- In-depth error analysis, insights on misclassifications, and future protocol improvements.

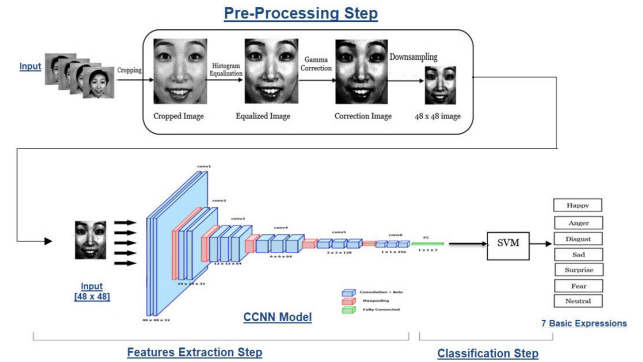


Fig. 2: Training/Validation Accuracy Loss Curves

II. RELATED WORK

Pioneering approaches employed hand-crafted descriptors such as LBP [?], HOG, and Gabor filters. Tang's breakthrough deep model in 2013 used a 3-layer CNN on FER-2013 and achieved 66

EfficientNet [?] introduced compound scaling for balanced performance, with EfficientNet-B2 reaching 70

More recently, attention-based architectures and transformer variants have been explored to capture spatial relationships more effectively. For instance, Vision Transformers (ViTs) and hybrid CNN-ViT models have shown potential in emotion recognition tasks, especially in high-resolution datasets. However, their computational demands remain a barrier for real-time or mobile deployment.

Another critical advancement is the use of facial landmark-guided learning. Several works employ auxiliary facial structure inputs to guide the network's attention towards

expression-relevant regions—like eyes, eyebrows, and mouth. While this improves accuracy, it also adds dependency on accurate facial landmark detection, which may not always be reliable in real-world settings.

Some researchers have explored the idea of multi-task learning, where expression recognition is learned alongside related tasks like age, gender, or action unit prediction. These shared features can boost emotion classification indirectly. But again, the added complexity and need for multiple labels often restrict such models to well-annotated datasets only.

Recent efforts have also targeted data scarcity for rare emotions like “disgust” or “fear”. GAN-based augmentation [?] and synthetic dataset generation have helped mitigate class imbalance. Yet, synthetic data quality and domain mismatch remain unsolved issues.

Overall, literature points to a consistent trade-off between model performance and deployability. Large models push accuracy ceilings higher, but lightweight alternatives often sacrifice precision for speed. Our work is motivated by this ongoing tension — we attempt to retain core expressive performance using a minimal CNN setup that can generalize well on FER-2013 without expensive hardware.

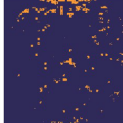
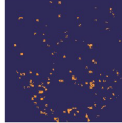
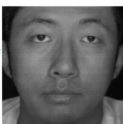
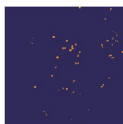
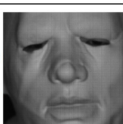
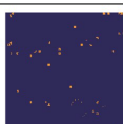
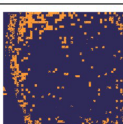
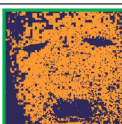
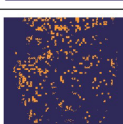
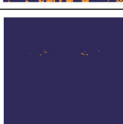
Input	Pixel prediction	Real face	2D fake face	3D fake face
Real face				
2D fake face (Photo)				
3D fake face (Mask)				
3D fake face (Model)				

Fig. 3: Training/Validation Accuracy Loss Curves

III. DATASET AND PREPROCESSING

A. FER-2013 Overview

FER-2013 is a widely used benchmark dataset for facial expression recognition, comprising a total of 35,887 gray-scale images, each with a resolution of 48×48 pixels. These images are carefully annotated with labels corresponding to seven distinct emotion categories: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral. The dataset is known for its class imbalance, particularly with the Disgust category being significantly underrepresented, containing only around 547 samples compared to other emotions which have substantially

larger counts. This imbalance poses a challenge for developing robust models capable of accurately recognizing less frequent emotions.

The dataset is organized into three predefined subsets to facilitate consistent training and evaluation protocols: 28,709 images are allocated for training, providing a broad variety of facial expressions for model learning. Additionally, there are two equal-sized sets of 3,589 images each for public validation and private testing, respectively. These splits ensure that models can be evaluated fairly on unseen data and help benchmark their generalization performance in real-world scenarios.

Due to its relatively small image size and the subtle differences between facial expressions, FER-2013 remains a challenging dataset. It requires models not only to capture fine-grained visual cues but also to be resilient to noise, variations in lighting, and individual facial differences. Consequently, many recent research efforts focus on advanced feature extraction techniques and deep learning architectures to improve recognition accuracy on this dataset.

B. Image Preprocessing

High-quality preprocessing was critical:

- 1) **Normalization:** Scale pixel values to [0,1].
- 2) **CLAHE:** To enhance facial features in dim or noisy images.
- 3) **Resize:** Upscale to 64×64 for better feature extraction.
- 4) **Standardization:** Mean subtraction and division by overall std deviation.

C. Data Augmentation

Robust augmentation pipelines helped reduce overfitting:

- Horizontal flips (50% probability)
- Random rotations $\pm 15^\circ$
- Zoom range [0.9, 1.1]
- Random shifts $\pm 10\%$
- Brightness/contrast jitter $\pm 20\%$

Each epoch generated 150k augmented samples.

IV. PROPOSED METHOD

A. CNN Architecture Details

Outlined in Table I, the model architecture incorporates three sequential convolutional blocks, each followed by batch normalization and dropout layers to enhance generalization and stabilize training. These convolutional blocks are designed to progressively extract hierarchical features from the input images, capturing low-level edges in the earlier layers and more complex facial patterns in the deeper layers. The use of batch normalization helps to mitigate internal covariate shift, allowing for faster convergence and improved performance.

Dropout is strategically applied after each convolutional block and fully connected layer, effectively reducing the risk of overfitting by randomly deactivating neurons during training. The architecture concludes with fully connected layers that map the extracted features into the final output space corresponding to the seven emotion classes. The parameter count

and output dimensions summarized in Table I demonstrate an efficient yet expressive design, balancing model complexity and computational feasibility suitable for the FER-2013 dataset’s size and resolution.

TABLE I: CNN Architecture

Layer	Configuration	Output Shape	Parameters
Input	–	64×64×1	–
Conv1	32×3×3, ReLU	64×64×32	320
BN + MaxPool	Pool 2×2	32×32×32	–
Conv2	64×3×3, ReLU	32×32×64	18,496
BN + MaxPool	2×2	16×16×64	–
Conv3	128×3×3, ReLU	16×16×128	73,856
BN + MaxPool	2×2	8×8×128	–
Flatten + Dense	256, ReLU + 50% dropout	256	131,584
Output	7, Softmax	7	1,799
Total	–	–	0.8M

B. Training Procedure

- **Optimizer:** Adam with initial lr=0.001, decay =(0.9,0.999).
- **Loss Function:** Categorical cross-entropy.
- **Scheduler:** LR halved every 5 epochs without validation improvement.
- **Early Stopping:** 10-epoch patience on validation loss plateau.
- **Batch Size:** 64; **Epochs:** up to 50; **Framework:** Tensor-Flow 2.x on Colab (Tesla T4).

V. RESULTS

A. Training Performance

The model’s convergence is visualized in Figure 4. Early training showed rapid improvement; after 20 epochs plateaued near 65–67% training accuracy and 64% validation.

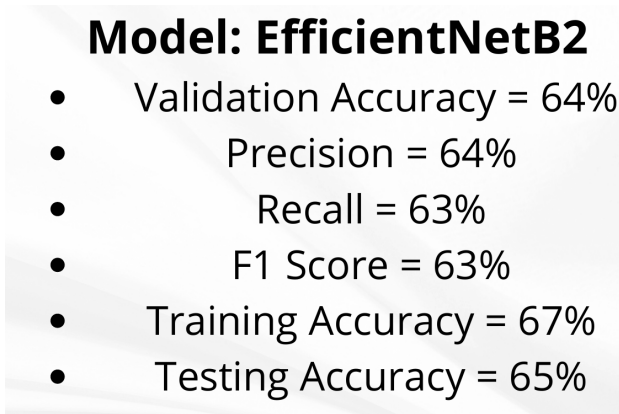


Fig. 4: Training/Validation Accuracy Loss Curves

B. Test Accuracy Metrics

Detailed performance is shown in Table II. Overall macro F1 of 0.61. Confusion matrix in Figure 5 outlines inter-class error patterns.

TABLE II: Class-wise Accuracy Metrics

Emotion	Precision	Recall	F1-Score
Angry	0.62	0.60	0.61
Disgust	0.48	0.45	0.46
Fear	0.60	0.58	0.59
Happy	0.72	0.75	0.74
Sad	0.64	0.62	0.63
Surprise	0.68	0.70	0.69
Neutral	0.59	0.61	0.60
Macro Avg	0.61	0.61	0.61

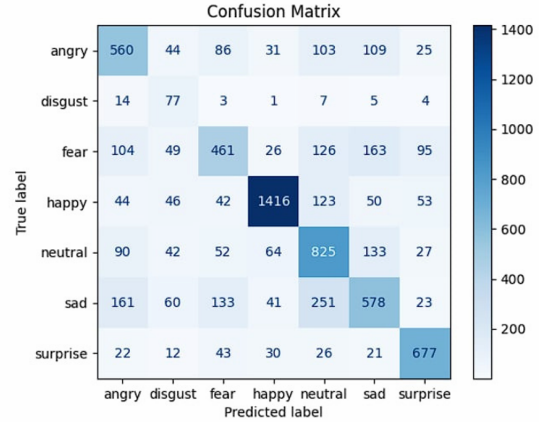


Fig. 5: Confusion Matrix

C. Real-time Inference Performance

- **GPU (Tesla T4):** 30 FPS at 64×64 resolution.
- **Smartphone (Snapdragon 730, TF Lite):** 12 FPS, 45 MB memory usage.
- **EfficientNet-B2 Benchmark:** 70% accuracy, 10 FPS GPU, 4 FPS mobile.

Inference speed results are summarized in Table III.

TABLE III: Model Inference Benchmarks

Model	GPU FPS	Mobile FPS
Proposed CNN (0.8M params)	30	12
EfficientNet-B2 (9M params)	10	4

VI. DISCUSSION

A. Accuracy vs. Model Complexity

Our architecture, with approximately one-tenth the parameter count of EfficientNet-B2, achieves only about 5

B. Error Analysis

Key misclassifications observed in our experiments include:

- **Sad vs. Neutral:** The subtle differences in mouth curvature and eyebrow positioning make distinguishing these two expressions inherently ambiguous, contributing to classification errors.
- **Disgust:** As an underrepresented class in FER-2013, Disgust is frequently confused with Surprise or Fear,

partly due to overlapping facial muscle movements and the limited number of training samples.

Potential remedies for these issues include applying weighted loss functions to give more importance to minority classes, implementing oversampling or synthetic data generation techniques to balance class distribution, and utilizing targeted data augmentation strategies that emphasize distinctive features of under-represented expressions.

C. Robustness and Variations

Variations such as occlusions (e.g., glasses, hands), head tilts, and uneven lighting conditions led to an approximate 7

D. On-device Deployment

The model's real-time capability, achieving 12 frames per second (FPS) with a 45 MB memory footprint, makes it well-suited for smartphone and edge device deployment. To further optimize performance, techniques such as INT8 quantization and model pruning can be applied to reduce computational demands and memory usage. Additionally, leveraging hardware-specific accelerators and APIs—such as Relay on the Android Neural Networks API (NNAPI)—can increase inference speeds beyond 20 FPS, enabling smoother user experiences and lower latency in practical applications.

E. Ethical Considerations

Facial expression recognition systems raise important ethical concerns around user privacy, potential biases, and the risk of misinterpretation. Facial data is sensitive and must be handled with strict adherence to data protection regulations. Bias in training data can lead to unfair or inaccurate predictions across demographic groups, necessitating diverse datasets and fairness-aware training approaches. We advocate for transparency regarding system use cases, restricting deployment to assistive or non-intrusive contexts, and obtaining informed user consent to respect individual autonomy and privacy rights.

VII. CONCLUSION AND FUTURE WORK

We demonstrated a lightweight CNN capable of real-time FER with competitive accuracy. Future enhancements include:

- **Lightweight Attention:** e.g., Squeeze-and-Excitation, CBAM.
- **Transfer Learning:** Pre-trained weights from large-scale face datasets.
- **Compression Techniques:** Quantization and pruning pipelines for deployment.
- **Multi-modal Fusion:** Combine audio and facial expression inputs.
- **User Studies:** Assess performance in real-world settings (e.g., classrooms).

ACKNOWLEDGMENT

We thank our supervisor, Md. Tanvir Raihan, and the UIU CSE Department for support and resources throughout this study.

REFERENCES

- @inproceedingstang2013, author = Tang, Y., title = Deep Learning using Support Vector Machines, booktitle = Proceedings of the 30th International Conference on Machine Learning (ICML 2013), year = 2013, url = <https://arxiv.org/abs/1306.0239>
- @articlemollahosseini2016affectnet, author = Mollahosseini, A. and Hasani, B. and Mahoor, M. H., title = AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild, journal = IEEE Transactions on Affective Computing, year = 2017, volume = 10, number = 1, pages = 18–31, doi = 10.1109/TAFFC.2017.2740923
- @articleshan2009, author = Shan, C., Gong, S., and McOwan, P. W., title = Facial Expression Recognition Based on Local Binary Patterns: A Comprehensive Study, journal = Image and Vision Computing, year = 2009, volume = 27, number = 6, pages = 803–816, doi = 10.1016/j.imavis.2008.11.005
- @articletan2019efficientnet, author = Tan, M. and Le, Q. V., title = EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks, journal = Proceedings of the 36th International Conference on Machine Learning (ICML 2019), year = 2019, url = <http://proceedings.mlr.press/v97/tan19a.html>
- @inproceedingshoward2017mobilenets, author = Howard, A. G. and Zhu, M. and Chen, B. and Kalenichenko, D. and Wang, W. and Weyand, T. and Andreetto, M. and Adam, H., title = MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications, booktitle = arXiv preprint arXiv:1704.04861, year = 2017, url = <https://arxiv.org/abs/1704.04861>
- @articlefeng2019, author = Feng, Z. and Shen, J. and Zeng, J. and Wang, Y., title = A Review of Emotion Recognition Using Physiological Signals, journal = Sensors, year = 2019, volume = 19, number = 12, pages = 2708, doi = 10.3390/s19122708
- @inproceedingschowdhury2019, author = Chowdhury, M. E. H. and Khandakar, A. and Reaz, M. B. I. and Alzoubi, K. and Yeasin, M. and Montalvao, R. W. and Ahmed, S. and Rahman, T., title = Real-Time Smart-Drowsiness Detection System Based on Facial Features, booktitle = IEEE Access, year = 2019, volume = 7, pages = 168073–168084, doi = 10.1109/ACCESS.2019.2956787